



LaTeX Formulas & Diagrams (Gantt Charts, ERDs & Flowcharts)

1. Overview

As teams scale on Confluence, documentation becomes harder to maintain - diagrams live in external tools, mathematical notation gets pasted as images and keeping technical docs up to date requires constant context switching.

This app solves that by bringing three professional visualization macros directly into Confluence:

- Create UML diagrams, flowcharts, sequence diagrams, ER diagrams, Gantt charts, and more - written as plain text, rendered instantly.
- Render display-style mathematical equations using LaTeX - centered, publication-quality, with a live preview as you type.
- Embed inline formulas inside paragraphs - formulas flow naturally within sentences, just like a LaTeX document.
- Access a built-in library of 80+ formula templates across 11 subject categories, from Basic Algebra to Maxwell's equations.
- Use all three macros on the same Confluence page - no external tools, no copy-paste, no lost context.

Designed for:

Engineering teams, technical writers, data scientists, educators, and researchers who need professional diagrams and mathematical notation inside Confluence - without leaving their documentation workflow.

2. What's Included

Easy UML Diagrams

Create any diagram type directly inside a Confluence page - write plain-text diagram syntax and see a live rendered preview as you type.

Includes:

- Flowchart, Sequence Diagram, Class Diagram, State Diagram, ER Diagram, Gantt Chart, Pie Chart, Git Graph, User Journey, Mind Map, Timeline, and Quadrant Chart all powered by Mermaid.
- Code editor on the left, rendered diagram preview on the right - updates in real time as you type.
- One-click access to ready-made examples for all 12 diagram types - load any sample into the editor instantly.
- Click and drag to navigate across large diagrams directly on the Confluence page.
- Copy the Mermaid syntax to clipboard directly from the editor.
- Clear syntax error display with expandable details so you can fix issues without guessing.

LaTeX Block Equations

Render display-style mathematical equations centered and full-width on your Confluence page - publication-quality output for technical and scientific documentation.

Includes:

- Formulas appear within flowing text rather than as centered blocks - the macro uses Confluence's inline layout so content wraps naturally around rendered math.
- Full formula library, live split-pane preview, multi-format input detection, and the three-engine rendering pipeline - identical to the block macro.
- All 11 subject categories and 80+ formulas available for inline use.
- Font size, theme, and color controls apply to inline output as well.

3. Use Cases

Diagrams & Technical Documentation

- System architecture diagrams - flowcharts and sequence diagrams embedded directly in architecture decision records.
- Database schema documentation - ER diagrams kept up to date inside the same Confluence page as the schema description.
- Project planning - Gantt charts created and edited in Confluence without exporting from external tools.
- Process documentation - state diagrams and user journey maps embedded in team runbooks and onboarding guides.
- Dependency mapping - Git graphs and relationship diagrams showing branch strategy or service dependencies.

Mathematical & Scientific Documentation

- Engineering specs - block equations for formulas, tolerances, and derived values embedded in design documents.
- Research notes - publication-quality LaTeX equations in Confluence wikis for academic or R&D teams.
- Data science documentation - statistical formulas, probability distributions, and model descriptions rendered inline.
- Physics and chemistry - Maxwell's equations, Schrödinger equation, chemical reaction notation via `\ce{}` support.
- Finance and analytics - mathematical formulas for scoring models, risk calculations, and metric definitions.

Mixed Content Documentation

- Technical tutorials - diagrams and inline math in the same page for courses, training materials, or knowledge bases.

- API and algorithm documentation - pseudocode flowcharts alongside formula definitions for algorithmic complexity.
- Sprint and delivery planning - Gantt charts and sequence diagrams embedded in planning pages updated each sprint.
- Stakeholder reports - professional diagram output and clean equation rendering that can be shared without screenshots.

4. Quick Start

Step 1: Add a Diagram

1. Open a Confluence page in edit mode.
2. Type / and search for Easy UML Diagrams → insert the macro.
3. In the editor, click Samples → pick a diagram type → the code loads automatically.
4. Edit the code as needed - the preview updates in real time.
5. Click Save & Publish. Use zoom, pan, and fullscreen controls on the published page.

Step 2: Add a Block Equation

1. Type / and search for LaTeX Block Equations → insert the macro.
2. Browse the formula library at the bottom - click a category tab, then click a formula card to load it.
3. Or type your own LaTeX, AsciiMath, or MathJS expression directly - the format is detected automatically.
4. Adjust font size and theme in the settings panel if needed.
5. Click Save - the equation renders centered and display-style on the page.

Step 3: Add Inline Math

1. Type / and search for LaTeX Inline Equations → insert the macro.
2. Write your paragraph text. Use $formula$ delimiters to embed formulas inline.
3. The preview shows the formula rendering inside the sentence in real time.
4. Click Save - the formula appears inline within the paragraph on the Confluence page.

5. Recommended Starter Setup

For a team getting started with the app:

Easy UML Diagrams

- Start with the Flowchart or Sequence Diagram samples - these cover the majority of technical documentation needs.
- Use the ER Diagram sample for database schema documentation; keep it on the same page as your schema description.
- Use the Gantt Chart for sprint or project planning pages - update the code directly in Confluence each sprint.
- For large diagrams, use the fullscreen and zoom controls in the published view rather than shrinking the diagram.

LaTeX Block Equations

- Start with the Physics or Algebra category in the formula library to get familiar with the editor.
- Use the Paper theme for documentation intended to be screenshot or shared - it produces clean parchment-style output.
- Turn on equation numbering for technical specs where equations are referenced later in the document.
- Use the PNG export if you need to include a formula in a presentation or external document.

LaTeX Inline Equations

- Use inline equations for definition paragraphs where a formula is part of the explanation, not a standalone element.
- Keep inline formulas short - single variables, simple expressions, or symbol definitions work best inline.
- Use block equations for anything complex - multi-line, fractions, matrices, or anything that needs visual prominence.

6. Key Benefits

Benefit	Description
Everything in Confluence	Diagrams, block equations, and inline math all live inside Confluence pages - no external tools, no tab switching, no stale screenshots.
Live preview as you type	Every keystroke updates the preview in real time across all three macros - what you see in the editor is exactly what renders on the page.
12 diagram types in one macro	Flowcharts, sequence diagrams, ER diagrams, Gantt charts, mindmaps, & seven more - all from the same Easy UML Diagrams macro with one-click samples.
Professional math notation	KaTeX renders LaTeX formulas at publication quality - the same engine used by academic publishers and research platforms.
80+ built-in formula templates	Formula library covers Basic, Algebra, Calculus, Statistics, Physics, Chemistry, and five more categories - each with live rendered previews.
Inline and block math	Block equations for display - style standalone formulas; inline equations for math embedded within paragraphs - both on the same page.
Interactive diagram controls	Fullscreen controls on every published diagram - no need to open a separate tool to review large or complex diagrams.
Zero infrastructure	Runs entirely on Atlassian Forge - no servers, no external databases, no data leaving your Atlassian environment.